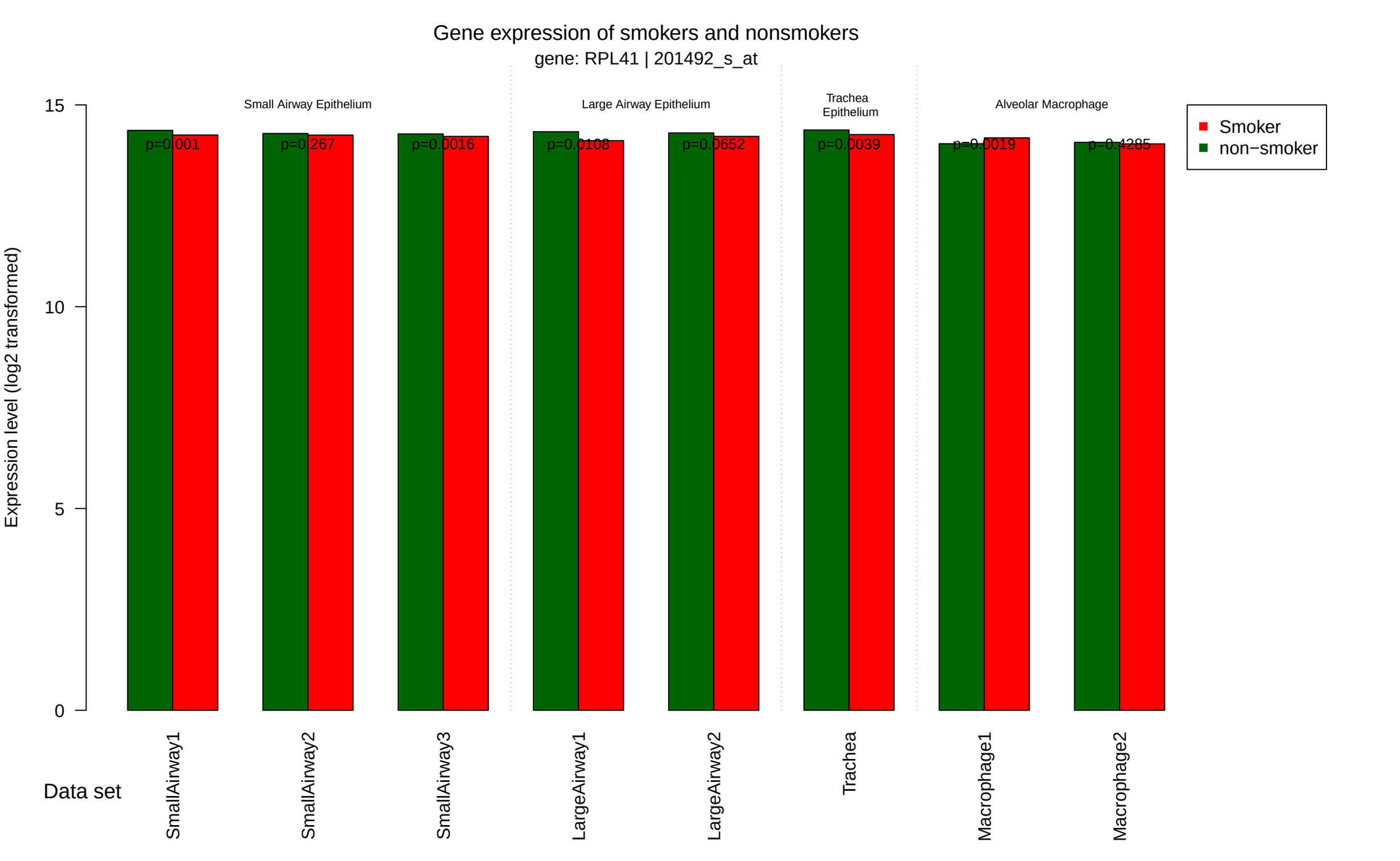
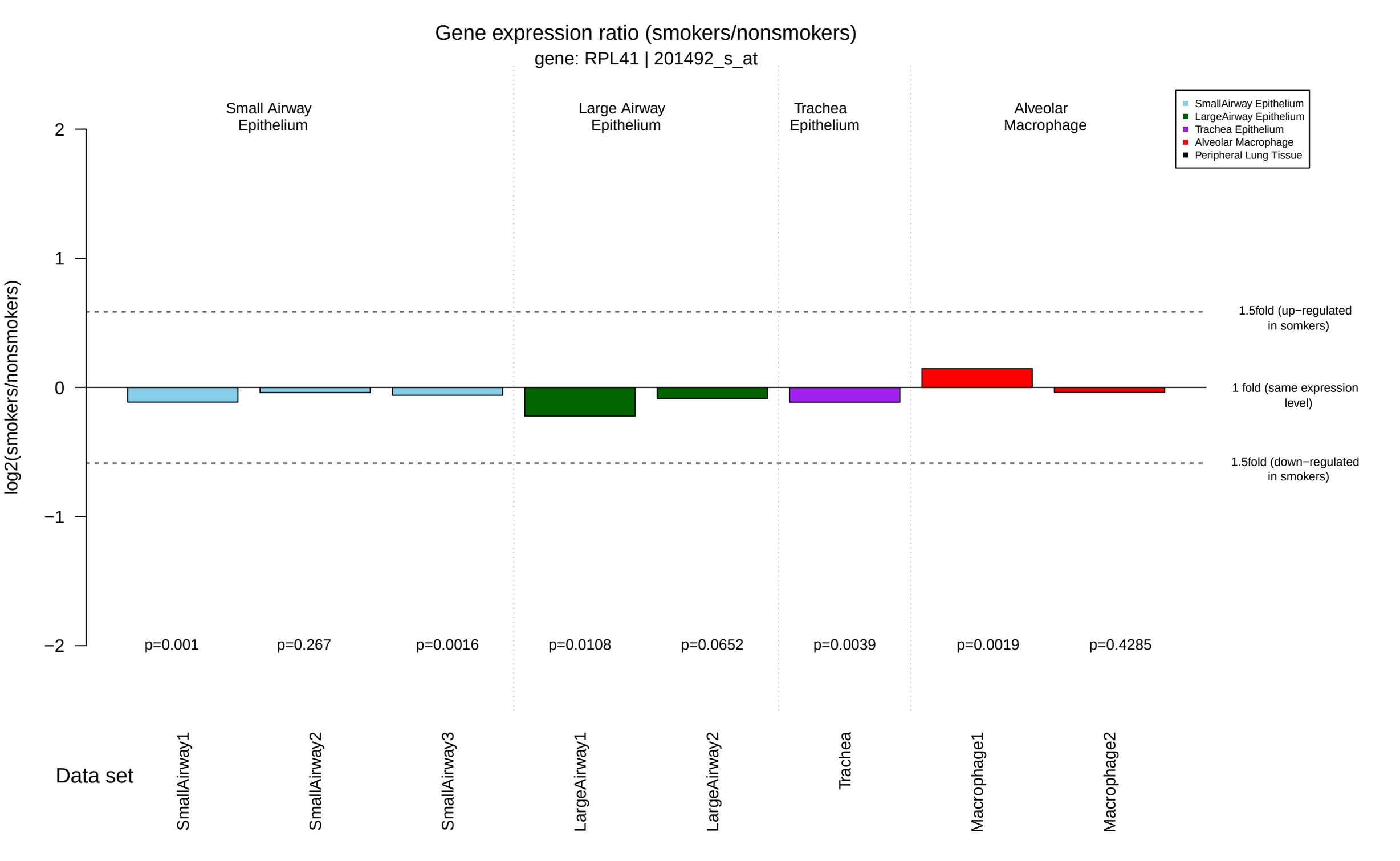
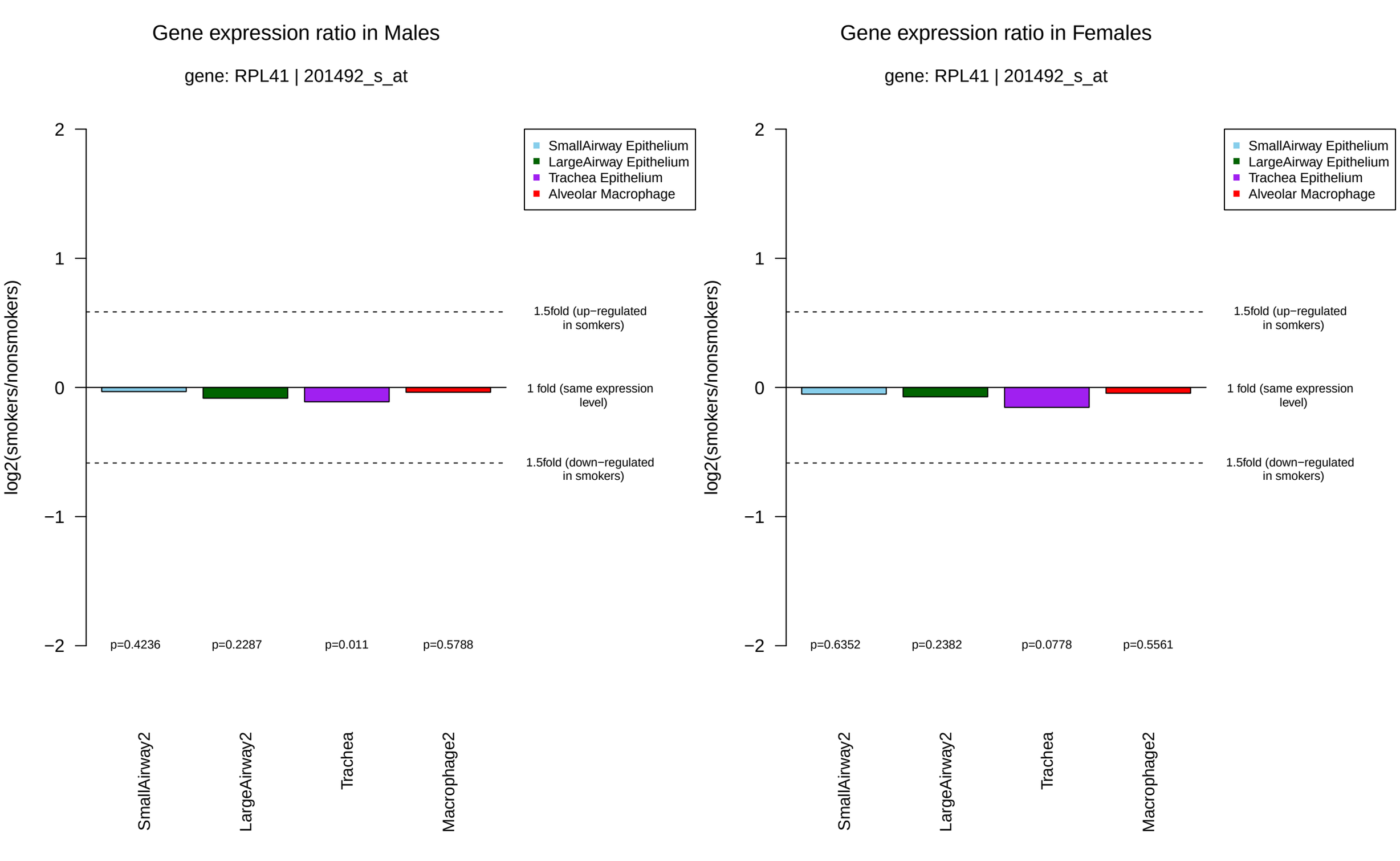
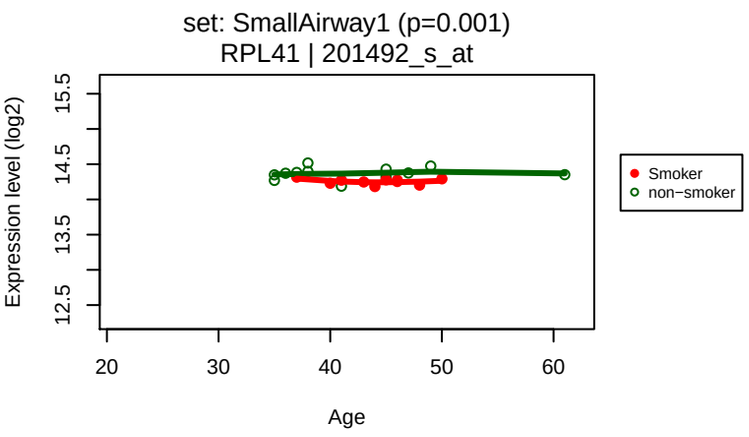
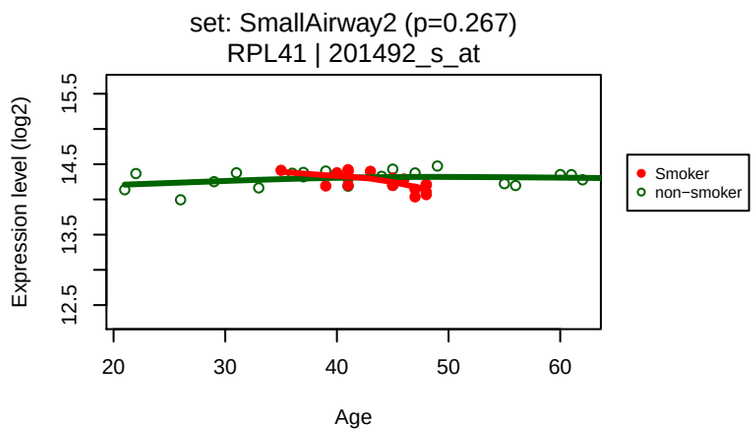
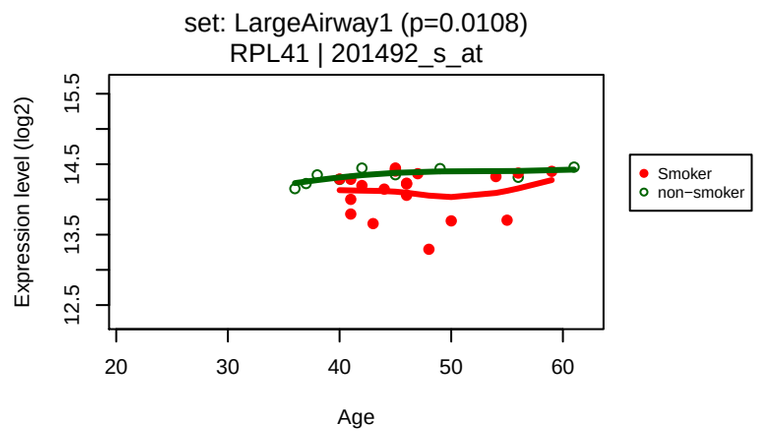

Gene expression of smokers and nonsmokers
gene: RPL41 | 201492_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
Smoker
non−smoker
p=0.001
p=0.267
p=0.0016
p=0.0108
p=0.0652
p=0.0039
p=0.0019
p=0.4285
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

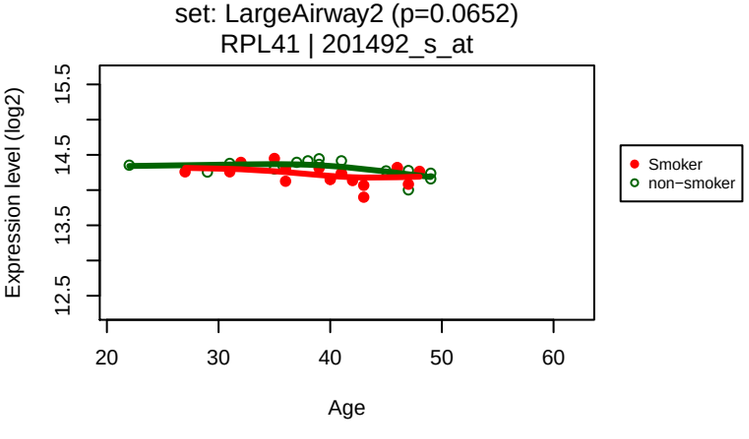

Gene expression ratio (smokers/nonsmokers)
gene: RPL41 | 201492_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.001
p=0.267
p=0.0016
p=0.0108
p=0.0652
p=0.0039
p=0.0019
p=0.4285
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

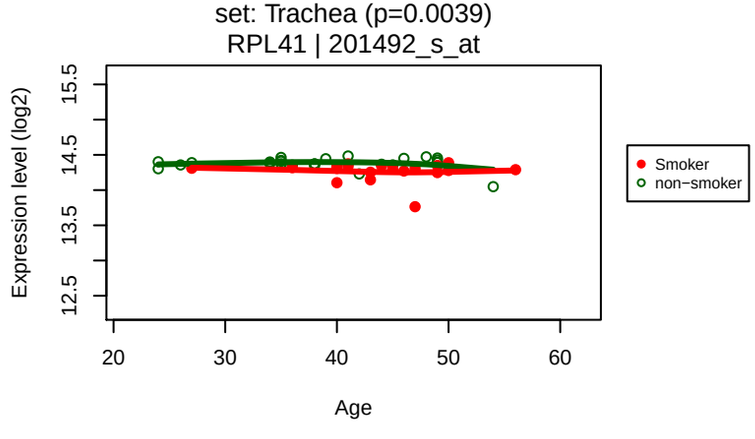

Gene expression ratio in Males
gene: RPL41 | 201492_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4236
p=0.2287
p=0.011
p=0.5788
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPL41 | 201492_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6352
p=0.2382
p=0.0778
p=0.5561
SmallAirway2
LargeAirway2
Trachea
Macrophage2

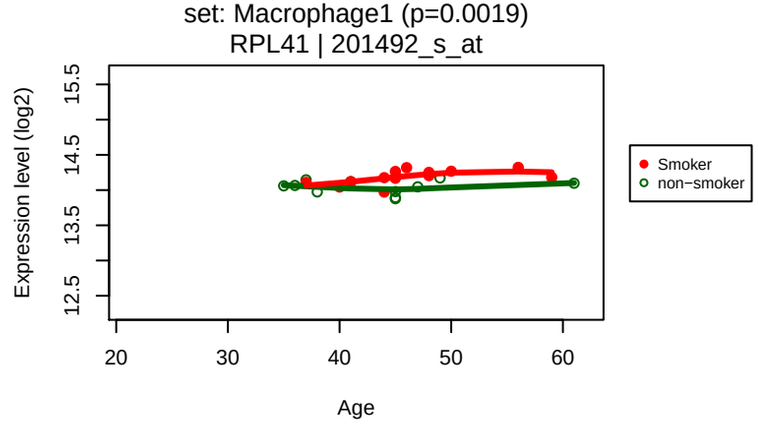

set: SmallAirway1 (p=0.001)
RPL41 | 201492_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker

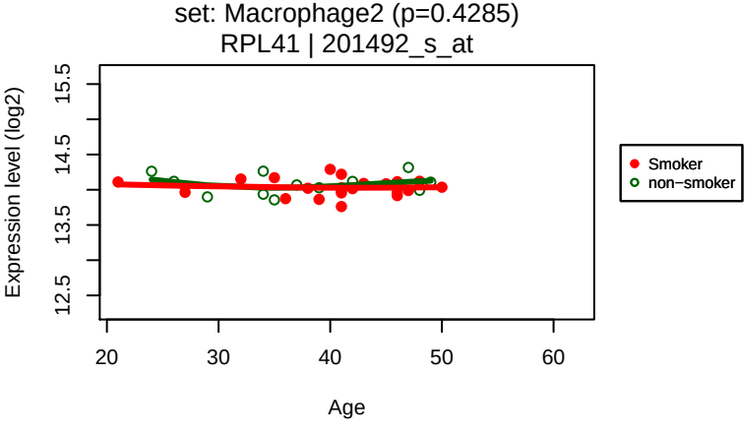

set: SmallAirway2 (p=0.267)
RPL41 | 201492_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker

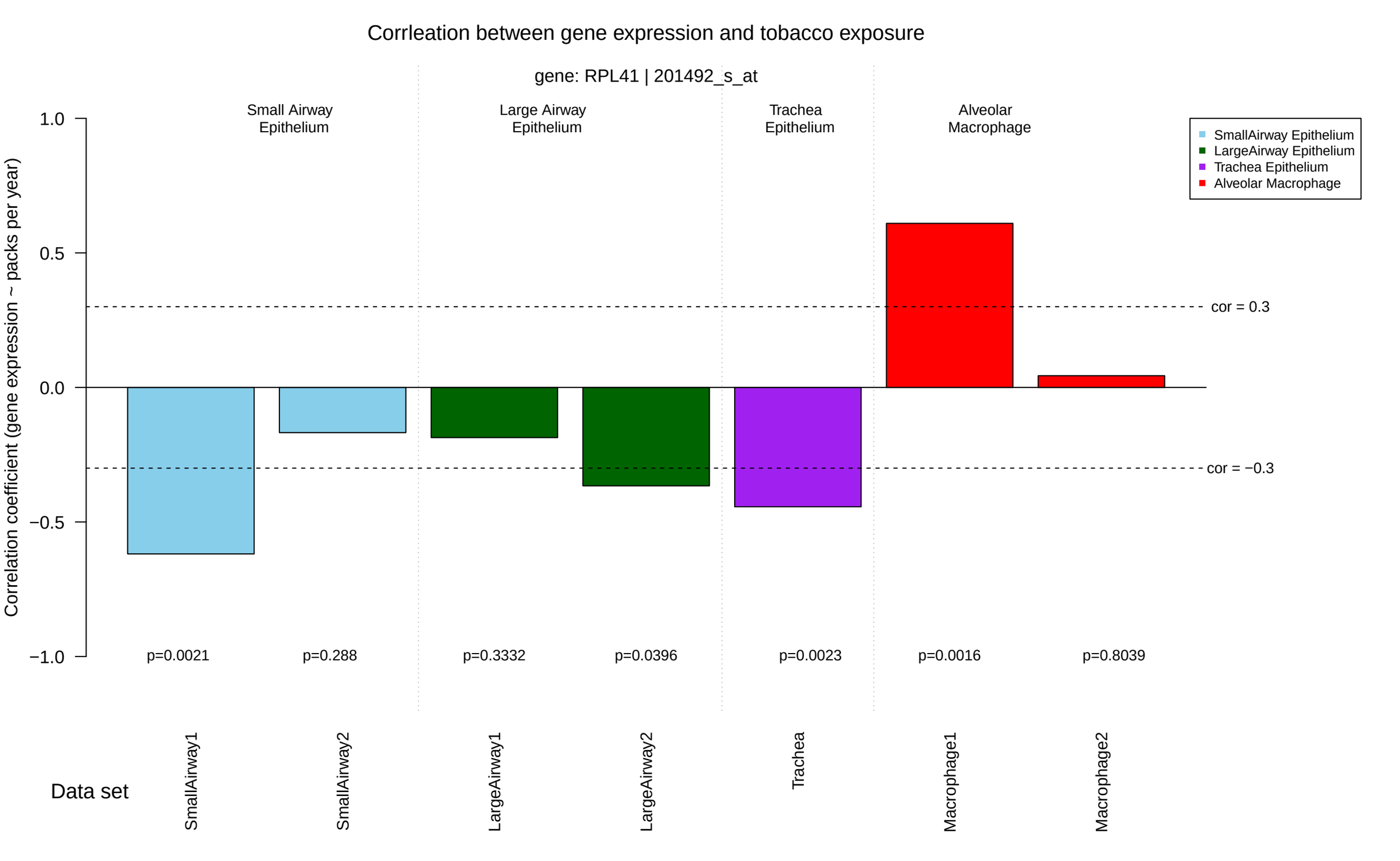

set: LargeAirway1 (p=0.0108)
RPL41 | 201492_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0652)
RPL41 | 201492_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0039)
RPL41 | 201492_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0019)
RPL41 | 201492_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4285)
RPL41 | 201492_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPL41 | 201492_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0021
p=0.288
p=0.3332
p=0.0396
p=0.0023
p=0.0016
p=0.8039
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2